

ESE 4170 - Final Project

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I. Introduction

Machine learning has become a central tool for extracting structure from data in domains where the relationship between measurable inputs and outcomes of interest is too complex to specify by hand. Supervised classification, in particular, addresses the problem of learning a mapping from feature vectors to discrete category labels using a set of labeled training examples. A well-trained classifier can then predict labels for new, unseen inputs. Among the most widely used classification algorithms are Support Vector Machines (SVM), which find a maximum-margin decision boundary in a (possibly kernel-induced) feature space; Artificial Neural Networks (ANN), which learn hierarchical nonlinear representations through layers of weighted connections; and Random Forests, which aggregate the predictions of many decorrelated decision trees to reduce variance and improve generalization.

In this project, we apply these three methods to the UCI Red Wine Quality dataset (Cortez et al., 2009), which contains 1,599 samples of Portuguese "Vinho Verde" red wine. Each sample is described by 11 physicochemical measurements obtained from objective laboratory tests. These measurements together make up a quality score between 0 and 10 assigned by professional wine tasters as the median of at least three independent evaluations. In this project, the original wine quality scores from 3 to 8 are used as class labels, so the task is treated as a multi-class classification problem.

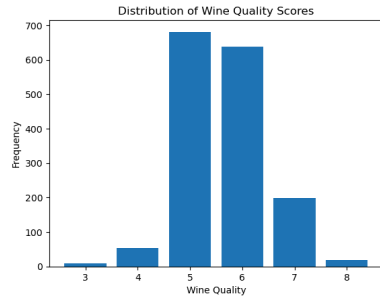
The goals of this project are threefold. First, we aim to build accurate predictors of perceived wine quality from purely physicochemical inputs, which would be of practical value for quality control applications where expert sensory evaluation is costly. Second, we use the comparison across SVM, ANN, and Random Forest to examine how different model families handle the structure of this dataset—particularly its class imbalance, its modest size, and the moderate correlations among input features. Third, we identify which physicochemical properties most strongly influence the model's predictions, providing some insight into the relationship between wine chemistry and perceived quality.

To achieve these goals, we adopt a common evaluation protocol across all three methods. Inputs are standardized where appropriate, the data are partitioned into stratified training and test sets, and hyperparameters are selected by cross-validated grid search on the training partition. Across all three methods, the models achieve reasonable performance on the multi-class classification task, particularly on the majority classes (quality scores 5 and 6), while struggling with minority classes due to class imbalance. Among the three methods, Random Forest achieved the highest overall performance in the multi-class setting, while SVM and ANN showed similar but slightly lower accuracy. Random Forest also provided useful feature importance insights, making it the most balanced model in this project.

II. Exploratory Analysis

1. Quality Score Distribution

First, we plotted the frequency distribution of quality scores. Scores 5 and 6 together make up most of the samples (681 and 638, respectively). 10 wines are rated 3, 53 wines are rated 4, and only 18 are rated 8.

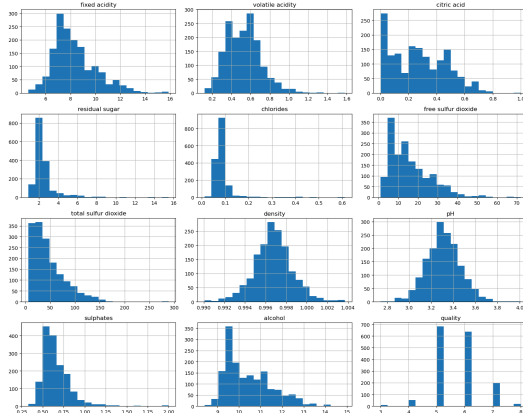


```
quality
3      10
4      53
5     681
6     638
7     199
8       18
Name: count, dtype: int64

Class imbalance ratio: 68.1x
```

2. Descriptive Statistics & Feature Distributions

Then, we ran comprehensive summary statistics and plotted histograms for all 12 variables.



	count	mean	std	min	25%	50%	75%	max
fixed acidity	1599.0	8.320	1.741	4.600	7.100	7.900	9.200	15.900
volatile acidity	1599.0	0.528	0.179	0.120	0.390	0.520	0.640	1.580
citric acid	1599.0	0.271	0.195	0.000	0.090	0.260	0.420	1.000
residual sugar	1599.0	2.539	1.410	0.900	1.900	2.200	2.600	15.500
chlorides	1599.0	0.087	0.047	0.012	0.070	0.079	0.090	0.611
free sulfur dioxide	1599.0	15.875	10.460	1.000	7.000	14.000	21.000	72.000
total sulfur dioxide	1599.0	46.468	32.895	6.000	22.000	38.000	62.000	289.000
density	1599.0	0.997	0.002	0.990	0.996	0.997	0.998	1.004
pH	1599.0	3.311	0.154	2.740	3.210	3.310	3.400	4.010
sulphates	1599.0	0.658	0.170	0.330	0.550	0.620	0.730	2.000
alcohol	1599.0	10.423	1.066	8.400	9.500	10.200	11.100	14.900
quality	1599.0	5.636	0.808	3.000	5.000	6.000	6.000	8.000

Fixed acidity, volatile acidity, citric acid, pH: Looking at features related to acidity, fixed acidity shows a moderate spread (mean 8.32, std 1.74). Volatile acidity is much tighter (mean 0.528, std 0.179). Based on the histograms, both fixed acidity and volatile acidity are roughly bell-shaped. Citric acid ranges from 0.0 to 1.0, with its histogram confirming that most wines have little to none (mean 0.271, std 0.195). pH is the most stable measure (mean 3.311, std 0.154) and is closest to a true normal distribution.

Free sulfur dioxide, total sulfur dioxide: Free sulfur dioxide has a wide range of 1.0 to 72.0, and total sulfur dioxide ranges even more from 6.0 to 289.0. Both have standard deviations large relative to their means. Both features are strongly right-skewed and have extremely large outliers. Free sulfur dioxide is heavily concentrated below 20. Total sulfur dioxide similarly peaks below 50 but stretches all the way to 289.

Residual sugar, chlorides, sulphates, alcohol, density: Residual sugar and Chlorides are both right-skewed. Both features have the majority of data packed in the low range, with large values forming a right tail. Density is tightly concentrated (mean 0.997, std 0.002), with its histogram close to a normal distribution. Alcohol has a relatively right-skewed distribution, peaking around 9.5 and spreading out to 14.9. Sulphates is moderately right-skewed, mostly concentrated between 0.4 and 0.8.

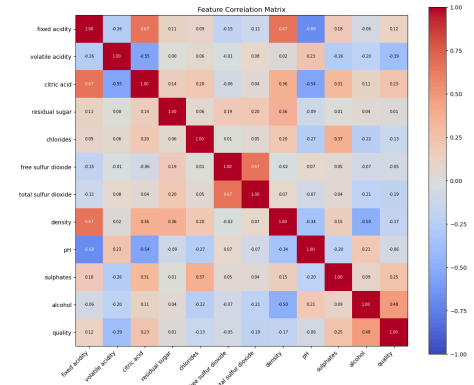
Quality: Quality scores range from 3 to 8 (mean 5.64, std 0.81). The histogram shows that most wines cluster around scores 5 and 6, with very few scoring below 4 and above 7.

3. Feature Correlation

We computed a correlation matrix across all features and visualized it as a heatmap. This helped us identify which features are most predictive of quality.

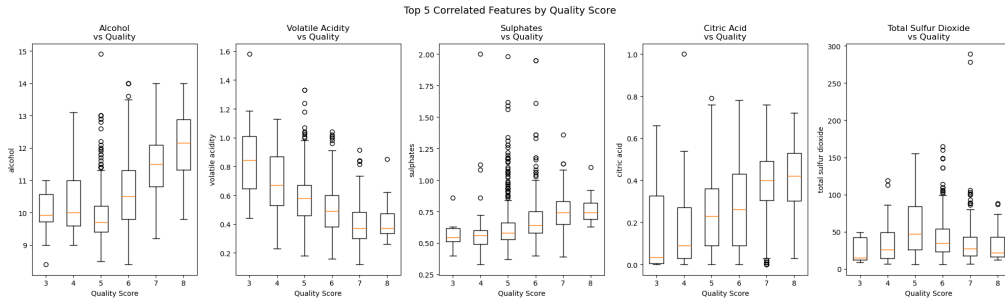
Alcohol came out as the strongest positive predictor (+0.48), followed by sulphates (+0.25) and citric acid (+0.23). **On the negative side, volatile acidity is the strongest predictor (-0.39)**, followed by total sulfur dioxide (-0.19) and density (-0.18). Residual sugar (+0.01) and free sulfur dioxide (-0.05) are essentially uncorrelated with quality on their own.

We also noticed some strong correlations between input features themselves. **Fixed acidity and citric acid (+0.67), fixed acidity and density (+0.67), and fixed acidity and pH (-0.68) are all highly correlated.** Free and total sulfur dioxide are also correlated (+0.67).



4. Top Feature Distributions and Outlier Detection

To better understand how the top-correlated features separate quality classes, we plotted boxplots for each of the five highest-correlated features grouped by quality score: **alcohol, volatile acidity, sulphates, citric acid, and total sulfur dioxide.**



Alcohol shows the clearest trend. Median alcohol rises consistently from ~10 at quality 3 to ~12 at quality 8, with relatively tight IQRs at the high end. **Volatile acidity also shows a clean trend that goes in the opposite direction**, which lines up with its negative correlation. Sulphates and citric acid both trend upward with quality, but with some overlap between categories. Total sulfur dioxide does not have a definitive trend, as it increases to quality 5, then decreases. Overall, these plots confirm that no single feature cleanly separates all quality classes on its own.

III. Methods

1. Support Vector Machine method

Support Vector Machines find decision boundaries that maximize the margin between classes in a feature space induced by a kernel function. Although SVM is originally a binary classifier, scikit-learn's SVC supports multi-class classification internally by combining multiple binary classifiers. In this project, we used the original wine quality scores from 3 to 8 as the class labels.

We implemented the SVM using scikit-learn's SVC inside a Pipeline with StandardScaler. Standardization was necessary because SVM is sensitive to feature scale, and the wine features have different units and

magnitudes. Using a pipeline also ensured that scaling was fit only on the training data during cross-validation, preventing data leakage.

To address class imbalance, we used `class_weight='balanced'`, which gives higher weight to minority classes. Hyperparameters were selected using GridSearchCV with 5-fold cross-validation on the training set. The grid searched over C, kernel type, and gamma, and weighted F1-score was used as the scoring metric because the task is imbalanced multi-class classification. The best model used an RBF kernel with $C = 5$ and $\gamma = 1$.

2. Artificial Neural Networks method

ANN is a multi-layer perceptron, and we used it to classify wine quality scores by doing forward propagation to pass input features through layers of weighted connections, compute a prediction, and then update the weights through backpropagation.

Before training, we applied StandardScaler normalization since features have very different scales and units. Normalizing them ensures no single feature dominates the gradient updates during backpropagation. We used a stratified 67/33 train/test split to preserve the class distribution across both sets. We used ReLU as the activation function and SGD as the optimizer. We also applied L2 regularization to prevent overfitting by penalizing larger weights.

To select the best architecture, we trained single hidden layer networks with node counts of 10, 20, 50, 100, 150, and 200, and compared test accuracy for each. Test accuracy peaked at 100 hidden nodes (0.6231) and did not improve further, so we used 100 nodes as our baseline.

We then ran GridSearchCV with 5-fold cross-validation. We scored on weighted F1, and the best configuration was hidden layers (20, 10), $\alpha = 0.001$, and learning rate = 0.01.

3. Random Forest method

3.1 Random Forest Model

We used a Random Forest classifier to perform multi-class classification on the wine quality dataset. It is an ensemble method that constructs multiple decision trees using bootstrap sampling and random feature selection, and combines their predictions through majority voting. This approach improves generalization and reduces overfitting.

3.2 Dataset and Setup

The task was treated as a multi-class classification problem using the original quality labels. No missing values were found, and no feature scaling was applied since Random Forest is not sensitive to feature magnitude. The dataset was split into training and testing sets using an 80/20 ratio with stratified sampling to preserve class distribution.

3.3 Model Configuration

A baseline Random Forest model with 100 trees was first trained. To improve performance, hyperparameter tuning was performed using grid search with 5-fold cross-validation. The parameters tuned included the number of trees, tree depth, and minimum samples required for splitting and leaf nodes. The best model used 200 trees, no depth limit, and a minimum of 5 samples per split.

3.4 Evaluation Metrics

Model performance was evaluated using accuracy, weighted precision, weighted recall, and weighted F1-score. Weighted metrics were used to account for class imbalance.

IV. Results and Analysis

1. Support Vector Machine method

Five-fold cross-validated grid search selected the RBF kernel with $C = 5$ and $\gamma = 1$. The best cross-validation weighted F1-score was 0.632. On the held-out test set, the SVM achieved an accuracy of 0.630 and a weighted F1-score of 0.610.

Table 1. SVM per-class performance on the test set.

Class	Precision	Recall	F1-score	Support
3	0.00	0.00	0.00	2
4	0.00	0.00	0.00	13
5	0.67	0.74	0.7	170
6	0.59	0.66	0.62	160
7	0.69	0.40	0.51	50
8	0.33	0.20	0.25	5
weighted	0.61	0.63	0.61	400

The SVM performed best on the majority classes, especially quality scores 5 and 6. Quality 5 achieved an F1-score of 0.70, while quality 6 achieved an F1-score of 0.62. The model struggled with minority classes, especially quality scores 3 and 4, which were not correctly predicted in the test set. This reflects the strong class imbalance in the dataset.

The confusion matrix shows that most errors occurred between adjacent quality classes, especially between scores 5 and 6. This suggests that wines with neighboring quality ratings have similar physicochemical characteristics, making exact multi-class prediction difficult. This reflects the increased difficulty of multi-class classification, where the model must distinguish among several closely related quality levels.

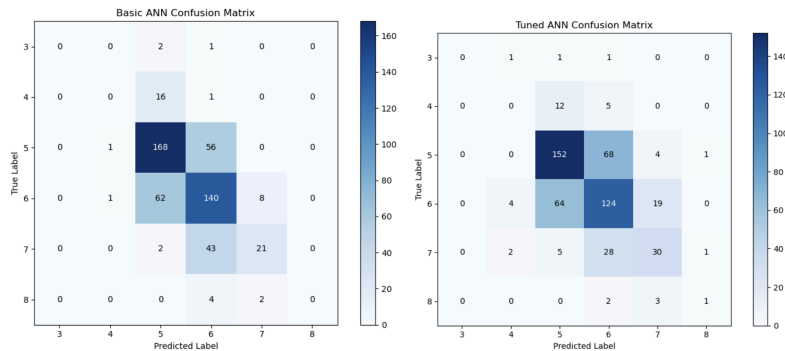
2. Artificial Neural Networks method

Model	Train Accuracy	Test Accuracy	Weighted Precision	Weighted Recall	Weighted F1-score
Basic ANN	0.677	0.623	0.599	0.623	0.601
Tuned ANN	0.789	0.581	0.565	0.581	0.572

The basic ANN achieved a test accuracy of 62.3% and a weighted F1 of 0.601. The tuned ANN scored lower on both test accuracy (58.1%) and weighted F1 (0.572), despite higher training accuracy (78.9%). This is largely attributable to the convergence warnings produced during GridSearchCV, where several configurations hit the maximum iteration limit before fully converging. Essentially, GridSearchCV selected the best parameters based on incompletely trained models. As a result, the “best” configuration found may not reflect the true optimal parameters. The higher training accuracy + lower test performance in the tuned ANN also suggests an overfitting issue to the training data without generalizing well.

Classification Report:					Best ANN Classification Report:				
	precision	recall	f1-score	support		precision	recall	f1-score	support
3	0.00	0.00	0.00	3	3	0.00	0.00	0.00	3
4	0.00	0.00	0.00	17	4	0.00	0.00	0.00	17
5	0.67	0.75	0.71	225	5	0.65	0.68	0.66	225
6	0.57	0.66	0.61	211	6	0.54	0.59	0.56	211
7	0.68	0.32	0.43	66	7	0.54	0.45	0.49	66
8	0.00	0.00	0.00	6	8	0.33	0.17	0.22	6
accuracy			0.62	528	accuracy			0.58	528
macro avg	0.32	0.29	0.29	528	macro avg	0.34	0.31	0.32	528
weighted avg	0.60	0.62	0.60	528	weighted avg	0.56	0.58	0.57	528

Looking at per-class performance, both models perform reasonably on quality 5 (F1 = 0.71 → 0.66) and quality 6 (F1 = 0.61 → 0.56). Both struggle significantly with minority classes. The tuned model does show improvement on quality 7 (F1 = 0.43 → 0.49) and quality 8 (F1 = 0.00 → 0.22). However, quality scores 3 and 4 are never predicted correctly in either model. This directly reflects the class imbalance, where scores 3 and 4 together make up less than 5% of the dataset.



The confusion matrices show that the most common errors in both models are adjacent-class misclassifications. Quality 5 was predicted as 6 (and vice versa), and quality 7 was frequently predicted as 6. This is expected given how concentrated the dataset is around scores 5 and 6.

3. Random Forest method

3.1 Overall Performance

The baseline model achieved an accuracy of 0.678 and a weighted F1-score of 0.663. After tuning, performance improved to an accuracy of 0.700 and a weighted F1-score of 0.684.

Model	Accuracy	F1-score(weighted)
Baseline	0.678	0.663
Tuned	0.700	0.684

3.2 Class-wise Analysis

The model performed best on the majority classes, such as quality 5 and 6, achieving F1-scores around 0.70. Performance on minority classes, such as quality 3, 4, and 8, was poor due to limited data, with near-zero recall for some classes. Confusion matrices show that most errors occur between adjacent classes, such as quality 5 and 6, indicating overlapping feature distributions.

3.3 Feature Importance

Feature importance analysis shows that alcohol is the most influential feature, followed by sulphates and sulfur dioxide. This suggests that chemical composition plays a key role in determining wine quality.

3.4 Discussion

Hyperparameter tuning improved performance by increasing model stability and reducing overfitting. However, class imbalance remains a major limitation, leading to poor performance on rare classes. Overall, Random Forest achieved solid performance but struggled with minority class prediction.

V. Conclusion

In this project, three classification methods—Support Vector Machine (SVM), Artificial Neural Networks (ANN), and Random Forest—were applied to the wine quality dataset and compared based on their performance.

The SVM model achieved a test accuracy of 63.0% and a weighted F1-score of 0.610 in the multi-class setting. It performed reasonably well on the majority classes, especially quality scores 5 and 6, but struggled with rare classes such as 3, 4, and 8.

The ANN models showed moderate performance, with the basic model achieving a test accuracy of 62.3% and a weighted F1-score of 0.601. The tuned ANN did not improve generalization and exhibited signs of overfitting, as indicated by higher training accuracy but lower test performance. This suggests that ANN models are sensitive to hyperparameter tuning and may require more careful optimization and training stability.

The Random Forest model achieved a balanced performance, with a final accuracy of 70.0% and a weighted F1-score of 0.684 in the multi-class setting. It performed well on the majority classes (quality scores 5 and 6) and provided interpretable feature importance results, identifying alcohol and sulphates as key predictors. However, like the other models, it struggled with minority classes due to severe class imbalance.

Overall, the results show that model performance is strongly influenced by class distribution. While SVM performed well on the majority classes, Random Forest provided a good balance between performance and interpretability, and ANN demonstrated the challenges of training more complex models on limited and imbalanced data.

Future improvements could include addressing class imbalance through resampling techniques, exploring more advanced models, and improving feature engineering to better separate adjacent quality classes.

Alice is responsible for the Introduction and SVM method. Christine is responsible for the EDA and ANN method. Twist is responsible for the Conclusion and Random Forest method. Everyone wrote section 3 and 4 on their own.

VI. Appendix